



EXPLORING AND EVALUATING THE CONCEPT OF PERSONALIZATION IN AUTONOMOUS DRIVING EXPERIENCES THROUGH MACHINE LEARNING

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AGENDA

1 INTRODUCTION

4 STUDY DESIGN

2 RELATED WORK

5 RESULTS

3 RESEARCH QUESTIONS

6 OUTLOOK

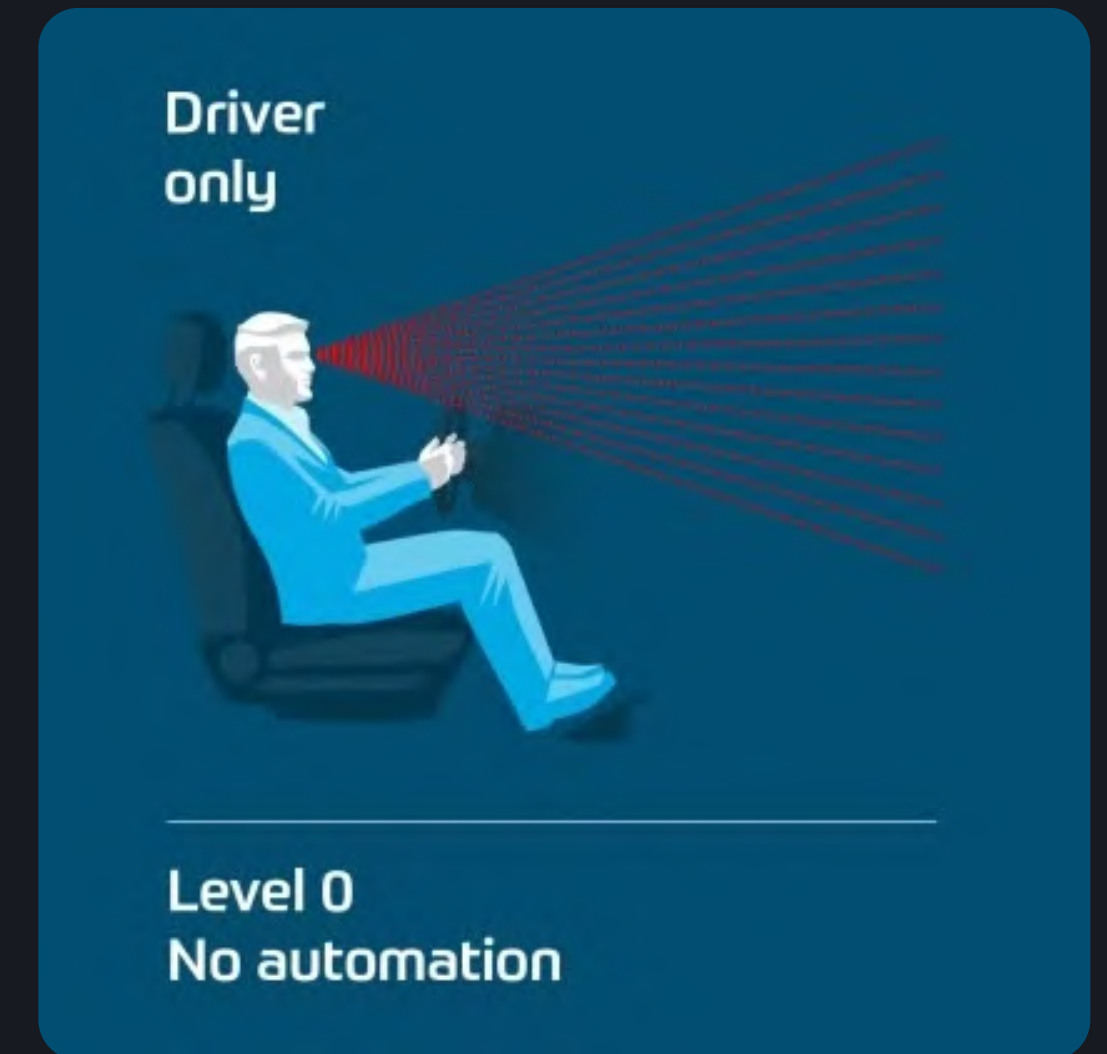


1. INTRODUCTION

SAE LEVELS

LEVEL 0: NO DRIVING AUTOMATION

- ❑ SAE Level 0: Non-automated vehicle.
- ❑ Driver: Full control over steering, braking, and accelerating.
- ❑ Assistance Systems: Minimal, only active in specific situations.
- ❑ Examples:
 - Electronic Stability Control (ESC)
 - Emergency Brake Assist (AEB)



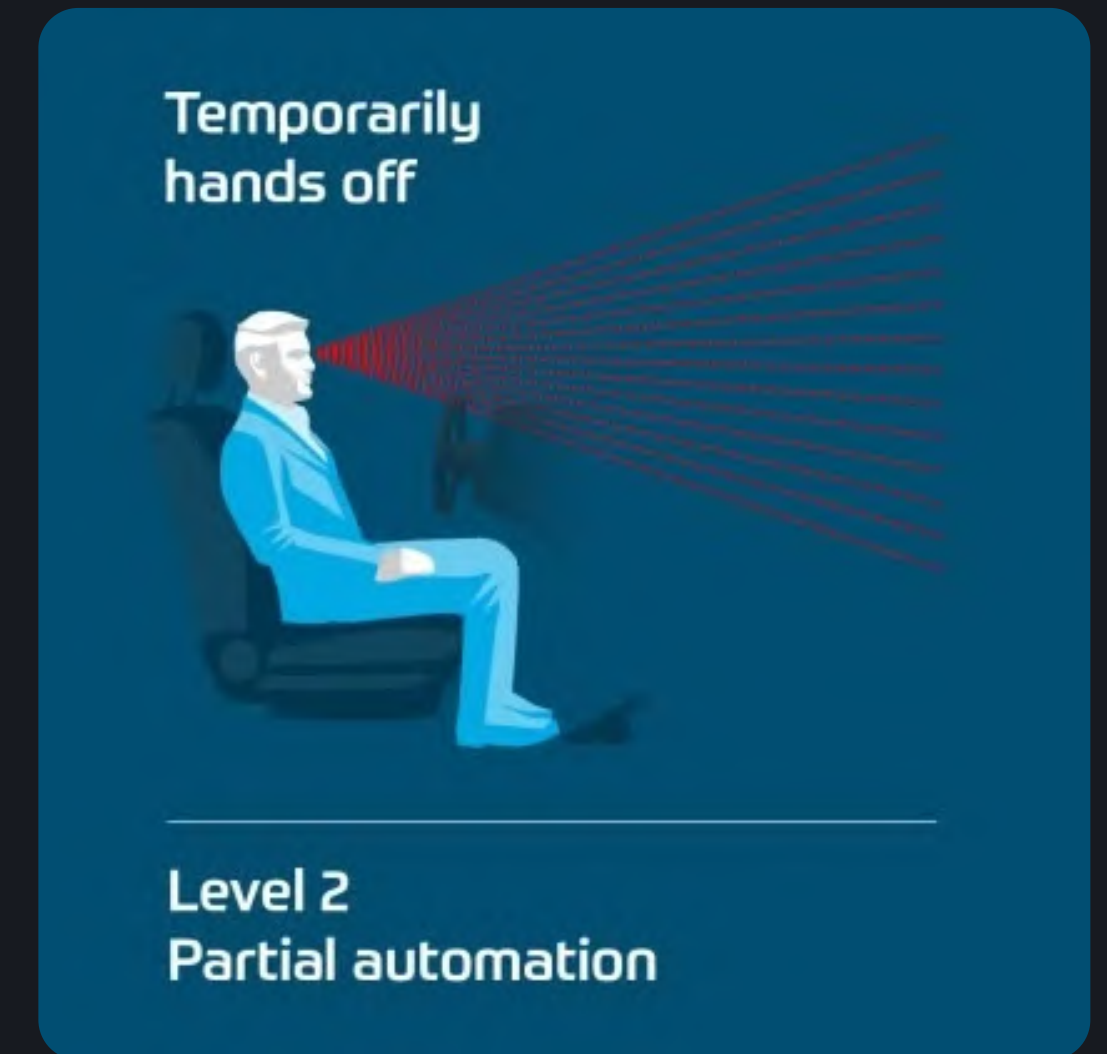
LEVEL 1 - DRIVER ASSISTANCE

- ❑ **Driver Role:** Fully responsible; must pay attention, steer, brake, and accelerate.
- ❑ **Assistance:** Supports either steering or braking/accelerating, but not simultaneously.
- ❑ **Examples:** Lane Keeping Assist (LKA), Adaptive Cruise Control (ACC).



LEVEL 2 - PARTIAL DRIVING AUTOMATION

- ❑ **Driver Role:** Must constantly monitor the system; touch the steering wheel regularly.
- ❑ **Assistance:** Controls both longitudinal and lateral dynamics simultaneously.
- ❑ **Examples:** Combined LKA and ACC, Level 2+ systems.
- ❑ **Restriction:** Writing messages on smartphones while driving is prohibited.



LEVEL 3 - CONDITIONAL DRIVING AUTOMATION

- ❑ **Driver Role:** May engage in non-driving activities within limits.
- ❑ **Assistance:** Vehicle temporarily takes over driving; driver must intervene after a warning period.
- ❑ **Examples:** Legal use of cell phones, watching movies in slow traffic or traffic jams.
- ❑ **Legal:** Germany approved Level 3 in 2017; Mercedes-Benz has Level 3 approval.

Temporarily
eyes off



Level 3
Conditional automation

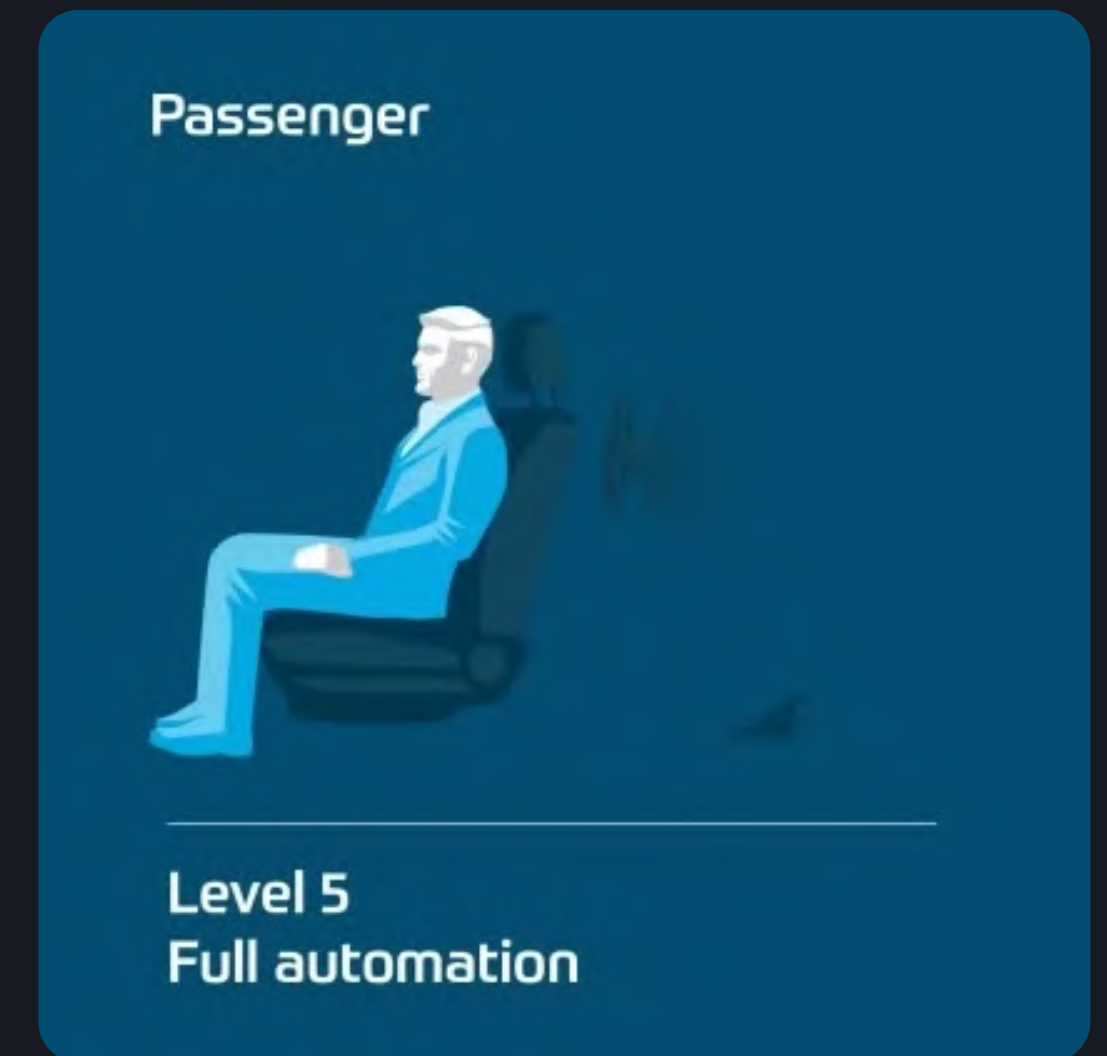
LEVEL 4 - HIGH DRIVING AUTOMATION

- ❑ **Driver Role:** No longer needs to be ready to intervene.
- ❑ **Assistance:** Vehicle operates autonomously under certain conditions (e.g., defined routes, highways).
- ❑ **Capabilities:** Vehicle can drive alone and reach a safe state without human intervention.



LEVEL 5 - FULL DRIVING AUTOMATION

- ❑ **Driver Role:** Does not exist; human is a passenger.
- ❑ **Assistance:** Vehicle acts completely autonomously in all conditions and environments.
- ❑ **Design:** No steering wheel, gas, or brake pedals required.



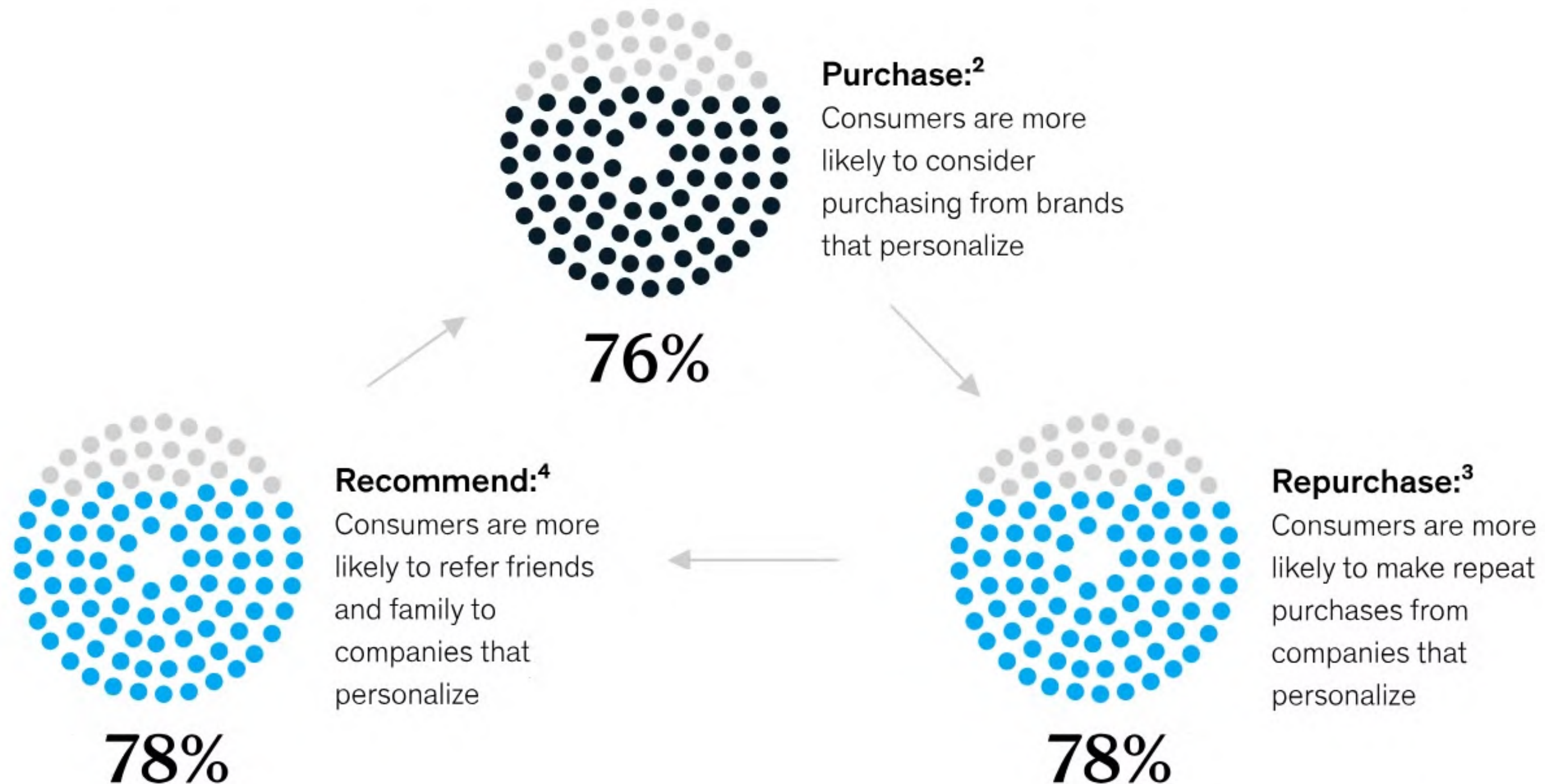
WHAT'S MISSING?



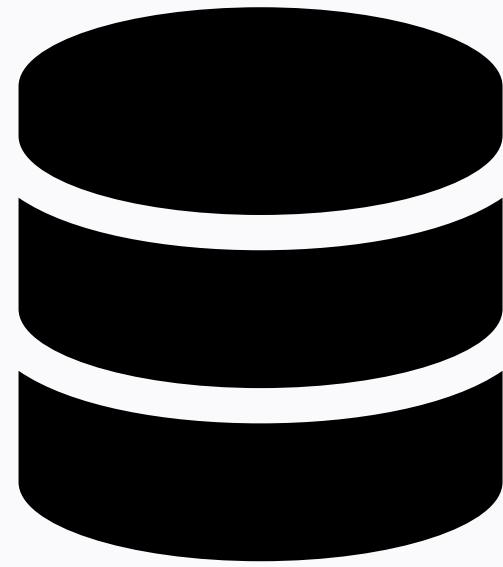
PERSONALIZATION

Personalization directly influences buying behavior across the customer life cycle.

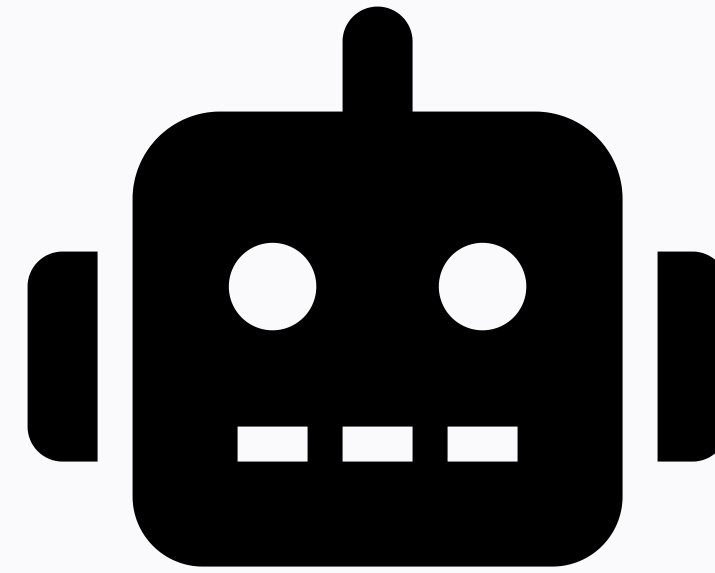
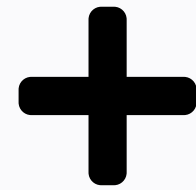
Likelihood to purchase, recommend, and repurchase depending on personalization, % of respondents¹



HOW A PRODUCT CAN BE **PERSONALIZED?**



DATA



MACHINE LEARNING



2. RELATED WORK

RESEARCH DIRECTIONS

- Exploring variables that affect the driving comfort, perceived safety, and trust [4][11].

- **Occupant Preference Metric** proposed to determine longitudinal & lateral value thresholds [4]

- Focusing on human comfort aspects to prevent motion sickness and provide **realism** [4]

- Different personality traits has effect on how individuals **trust** machines [3]

- Driving style differ greatly amongst individuals. [15]

- Personalizing the driving mode of an autonomous vehicle can improve the driver's comfort by reducing stress and increasing excitement. [17]

- **Personalization** may significantly increase trust, use, acceptance. [5]

- **Personality traits** play crucial role on new technology acceptance in automated driving [8]

LONGITUDINAL & LATERAL VALUES

Longitudinal
Acceleration

Lateral
Acceleration

Detect
Vehicle

Steering Action

Speed

Overtake

Jerk

Lane Offset

Deceleration

[4] [6] [11] [19]



3. RESEARCH QUESTIONS

RQ1

How can end-users' manipulation of selected variables in **automated vehicles** influence their **driving experience** and **satisfaction**?

RQ2

What impact does user customization of autonomous vehicle control parameters (such as longitudinal acceleration, deceleration, overtake etc.) have on **driving comfort** and **perceived safety**?

RQ3

What correlations exist between **user personality dimensions** and **preferences** for autonomous vehicle behavior **customization**?

RQ4

How does the ability to customize specific driving parameters in an autonomous vehicle affect **user trust** and **acceptance** of the technology?

RQ5

How do users perceive the difference between **automated driving** and **personalized, customizable automated driving** experiences?



4. STUDY DESIGN

BIG **5** - PERSONALITY QUESTIONNAIRE

EXTRAVERSION

- SOCIABILITY
- ASSERTIVENESS
- ENERGY

AGREEABLENESS

- COMPASSION
- COOPERATIVENESS
- TRUST

CONSCIENTIOUSNESS

- ORGANIZATION
- DEPENDABILITY
- DISCIPLINE

NEUROTICISM

- ANXIETY
- MOODINESS
- EMOTIONAL INSTABILITY

OPENNESS

- IMAGINATION
- CREATIVITY
- CURIOSITY

BIG **5** - PERSONALITY QUESTIONNAIRE

NEUROTICISM

- ANXIETY
- MOODINESS
- EMOTIONAL INSTABILITY

OPENNESS

- IMAGINATION
- CREATIVITY
- CURIOSITY

STUDY TASK SELECTION

- Questionnaire Participants = **57**
- Diverse Ethnicity, Background
- Freedom to choose participants
- **High Openness - Low Neuroticism**
- **High Neuroticism - Low Openness**

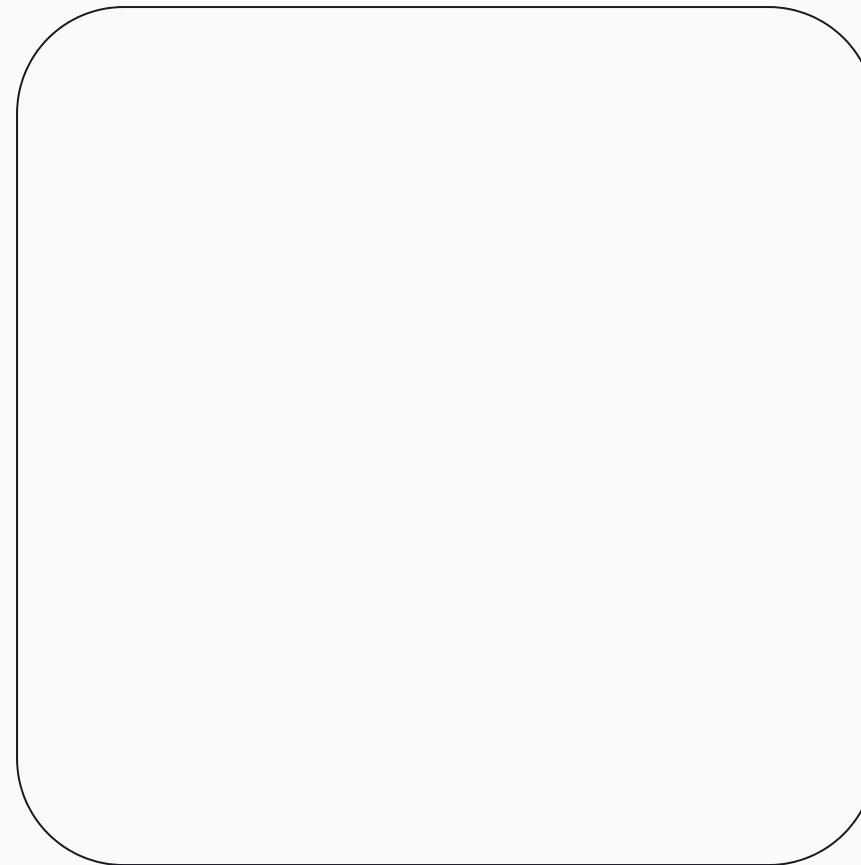
STUDY SETUP

- **Goal:** Explore Automated Driving customization based on personality trait
- **Duration:** 1 week
- **Participants:** 27
- **Personality Type 1:** High Openness - Low Neuroticism **13** P
- **Personality Type 2:** High Neuroticism - Low Openness **14** P

STUDY CONDITIONS



A: FUNCTIONAL UI



B: NO UI



C: PLACEBO UI

STUDY CONDITIONS

Mixed Factorial Design

Group 1

A: Functional UI

B: No UI

Group 2

A: Functional UI

C: Placebo UI

IV: Personality traits,
experiment condition

DV: UX, Acceptance, Trust,
Perceived Safety

QUANTITATIVE ANALYSIS

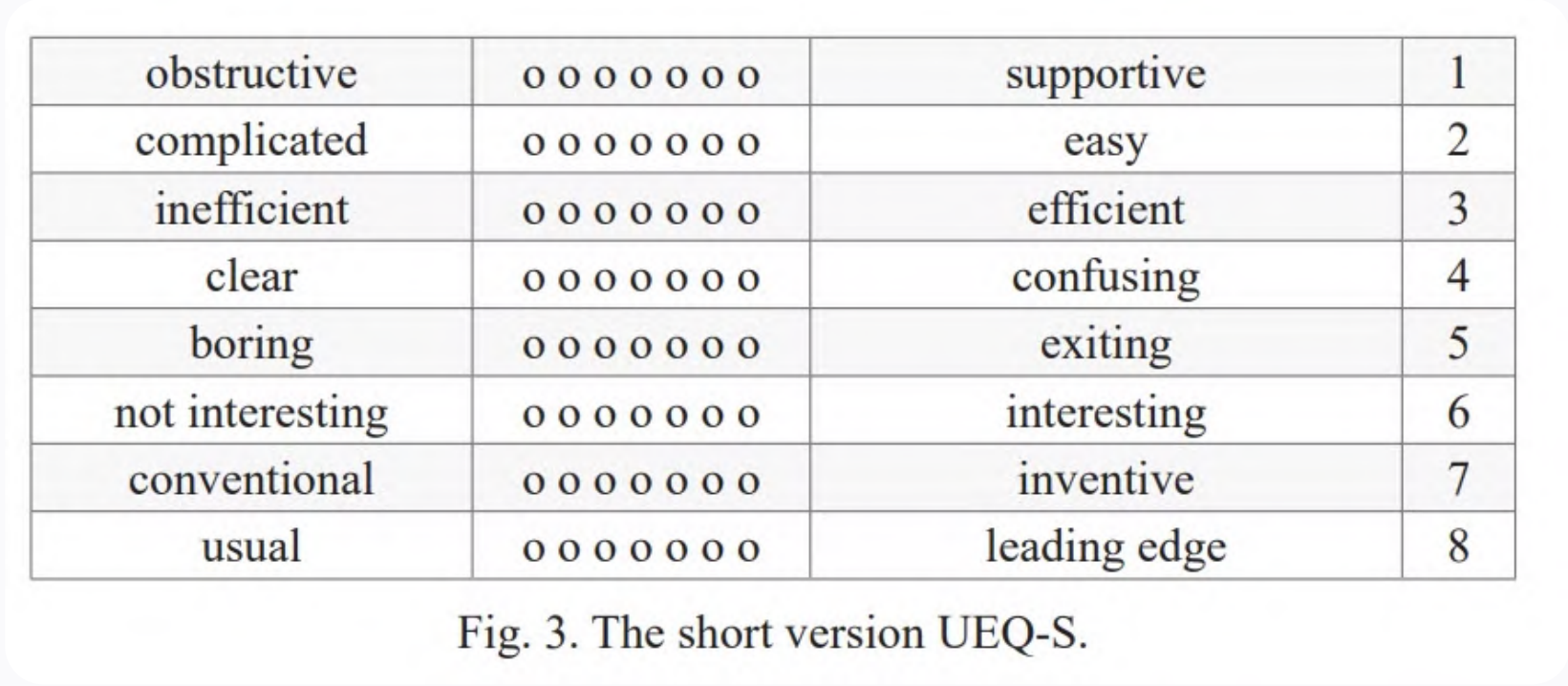


Table 1. The nine items in the questionnaire

My judgements of the (...) system are... (please tick a box on every line)

1	useful	_____	useless
2	pleasant	_____	unpleasant
3	bad	_____	good
4	nice	_____	annoying
5	effective	_____	superfluous
6	irritating	_____	likeable
7	assisting	_____	worthless
8	undesirable	_____	desirable
9	raising alertness	_____	sleep-inducing





WIP - MACHINE LEARNING

	A	B	C	D	E	F	G	H	I	J	K	L
1	Timestamp	Acceleration	Deceleration	Lane Offset	Consider Overtake	Overtake	Steer Trq	Steer Accel	Steer Veloc	Speed	Lateral Acceleration	Personality Trait
2	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	130.25	3.95	High Openness - Low Neuroticism
3	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	130.5	3.95	High Openness - Low Neuroticism
4	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	128.25	3.95	High Openness - Low Neuroticism
5	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	132.75	3.95	High Openness - Low Neuroticism
6	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	133	3.95	High Openness - Low Neuroticism
7	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	133	3.95	High Openness - Low Neuroticism
8	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	133.25	3.95	High Openness - Low Neuroticism
9	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	133.25	3.95	High Openness - Low Neuroticism
10	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	133.25	3.95	High Openness - Low Neuroticism
11	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	133.5	3.95	High Openness - Low Neuroticism
12	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	133.5	3.95	High Openness - Low Neuroticism
13	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	133.75	3.95	High Openness - Low Neuroticism
14	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	133.75	3.95	High Openness - Low Neuroticism
15	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	133.75	3.95	High Openness - Low Neuroticism
16	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	134	3.95	High Openness - Low Neuroticism
17	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	134	3.95	High Openness - Low Neuroticism
18	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	134.25	3.95	High Openness - Low Neuroticism
19	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	134.25	3.95	High Openness - Low Neuroticism
20	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	134.25	3.95	High Openness - Low Neuroticism
21	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	134.5	3.95	High Openness - Low Neuroticism
22	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	134.5	3.95	High Openness - Low Neuroticism
23	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	134.75	3.95	High Openness - Low Neuroticism
24	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	134.75	3.95	High Openness - Low Neuroticism
25	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	134.75	3.95	High Openness - Low Neuroticism
26	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	135	3.95	High Openness - Low Neuroticism
27	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	135	3.95	High Openness - Low Neuroticism
28	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	135.25	3.95	High Openness - Low Neuroticism
29	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	135.25	3.95	High Openness - Low Neuroticism
30	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	135.25	3.95	High Openness - Low Neuroticism
31	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	135.5	3.95	High Openness - Low Neuroticism
32	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	135.75	3.95	High Openness - Low Neuroticism
33	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	135.75	3.95	High Openness - Low Neuroticism
34	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	135.75	3.95	High Openness - Low Neuroticism
35	6/18/2024 11:16	5.0688	-4.8	0		0	0	10750	5	136	3.95	High Openness - Low Neuroticism

H1

The option to customize the automation's driving style has effects on participants' **user experience**.

H2

Participants that were able to customize will report higher driving comfort compared to other participants that didn't customize.

H3

Participants with **high openness** and **low neuroticism** will show a stronger preference for automated vehicle behavior customization compared to participants with **low openness** and **high neuroticism**.

H4

Participants in condition A (Functional UI) will report **higher satisfaction** and **usefulness** compared to participants in condition B (No UI) and C (Placebo UI)

H5

Option to customize the driving style has an effect on participants' **trust**.

”

*"Condition with control UI was
more comfy, no UI was **scary**
I prefer to have control on myself"*

”

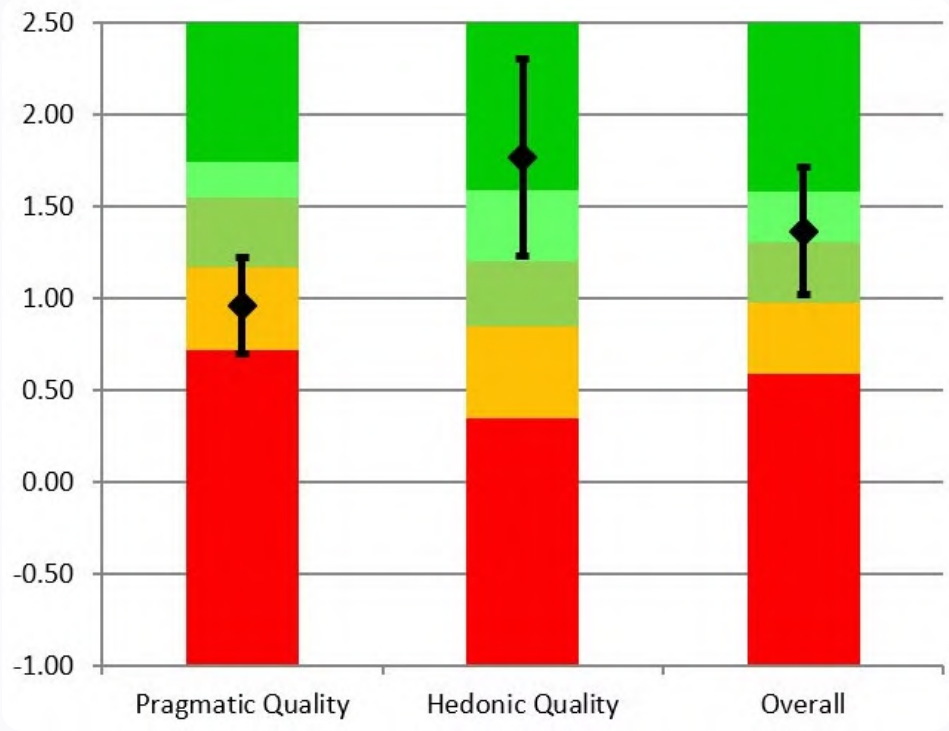
*"Personalized option was **better**,
No UI option felt **less safe** and had
faster turns, it was not comfortable"*



5. RESULTS

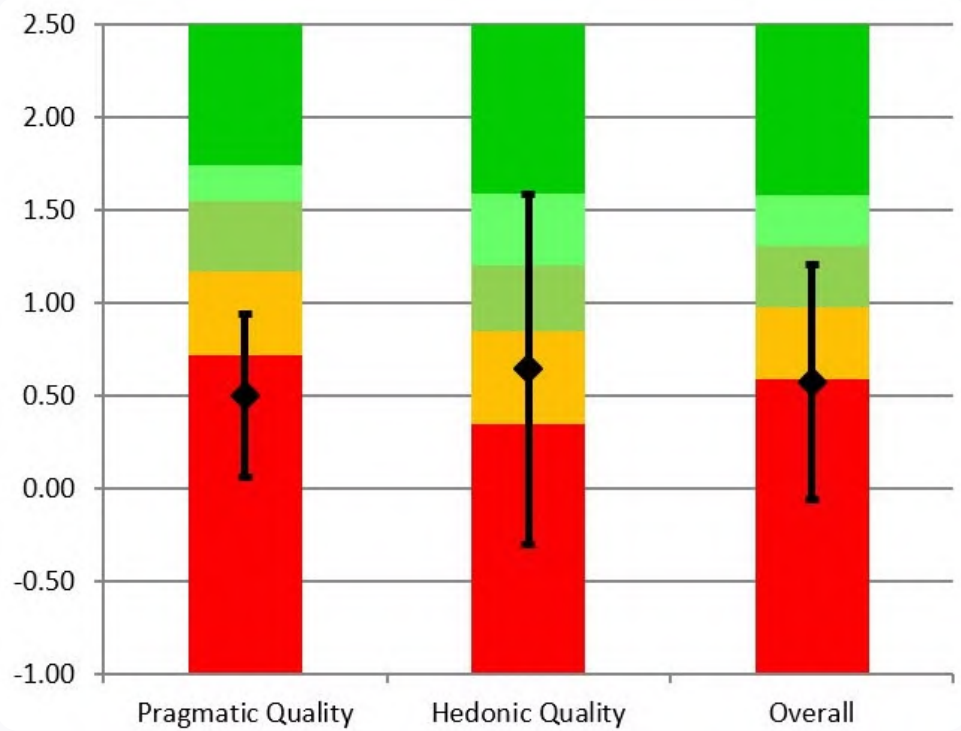
UEQ-S RESULTS

Functional UI (A)



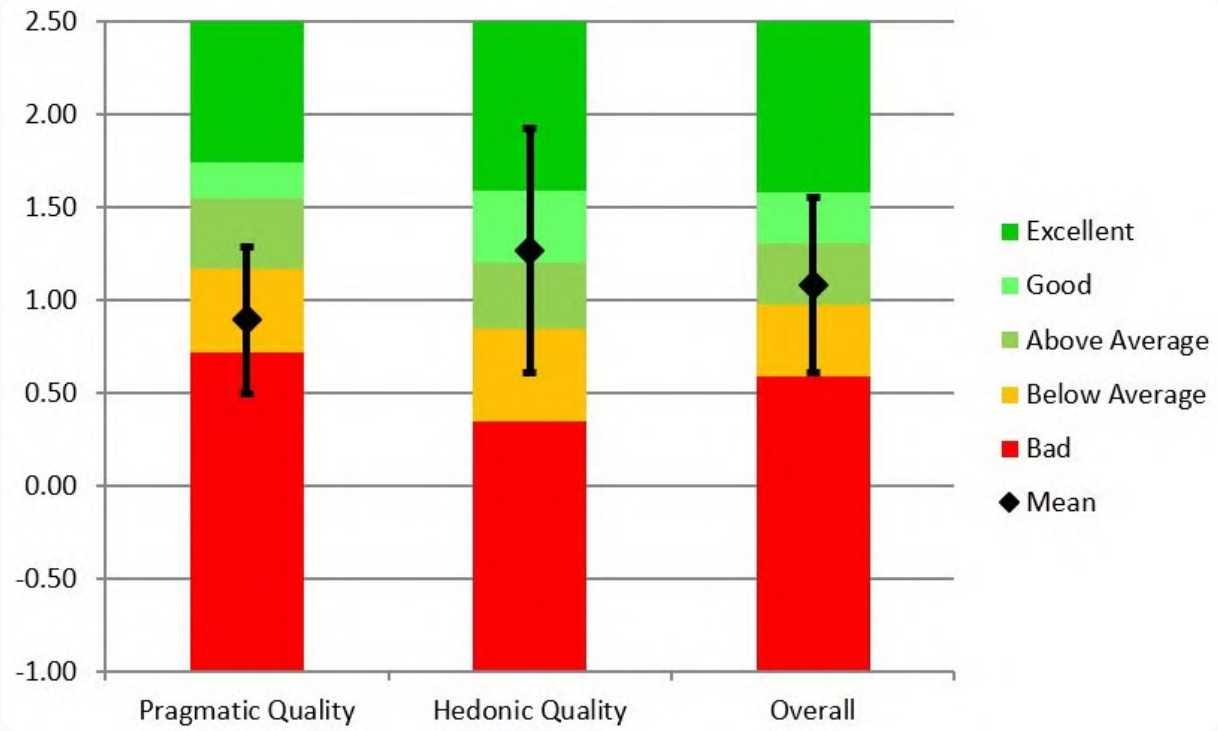
Scale	Mean	Comparisson to benchmark
Pragmatic Quality	0.961538462	Below average
Hedonic Quality	1.769230769	Excellent
Overall	1.37	Good

No UI (B)



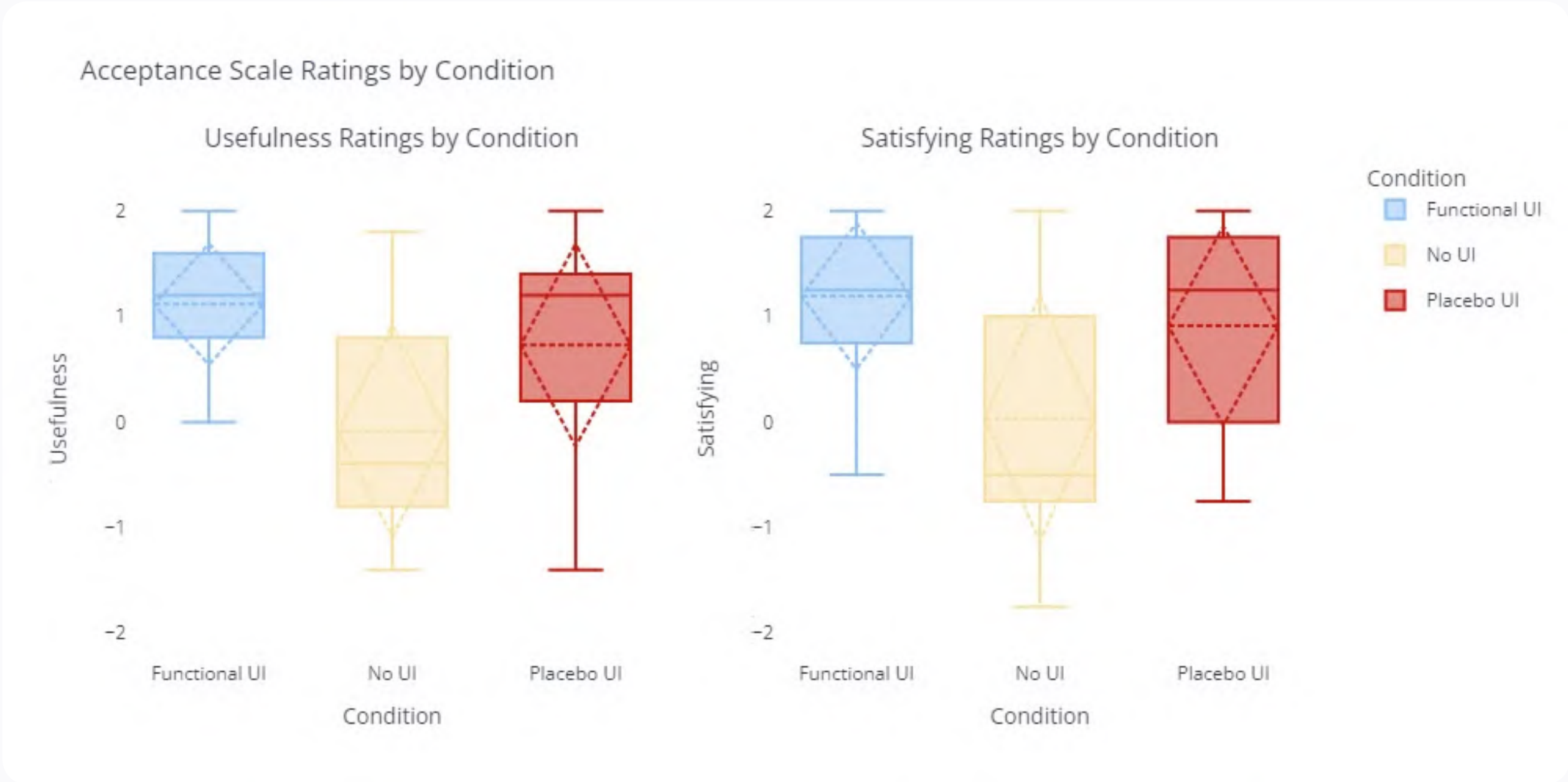
Scale	Mean	Comparisson to benchmark
Pragmatic Quality	0.5	Bad
Hedonic Quality	0.642857143	Below Average
Overall	0.57	Bad

Placebo UI (C)



Scale	Mean	Comparisson to benchmark
Pragmatic Quality	0.892857143	Below average
Hedonic Quality	1.267857143	Good
Overall	1.08	Above Average

ACCEPTANCE SCALE



Usefulness Metrics:

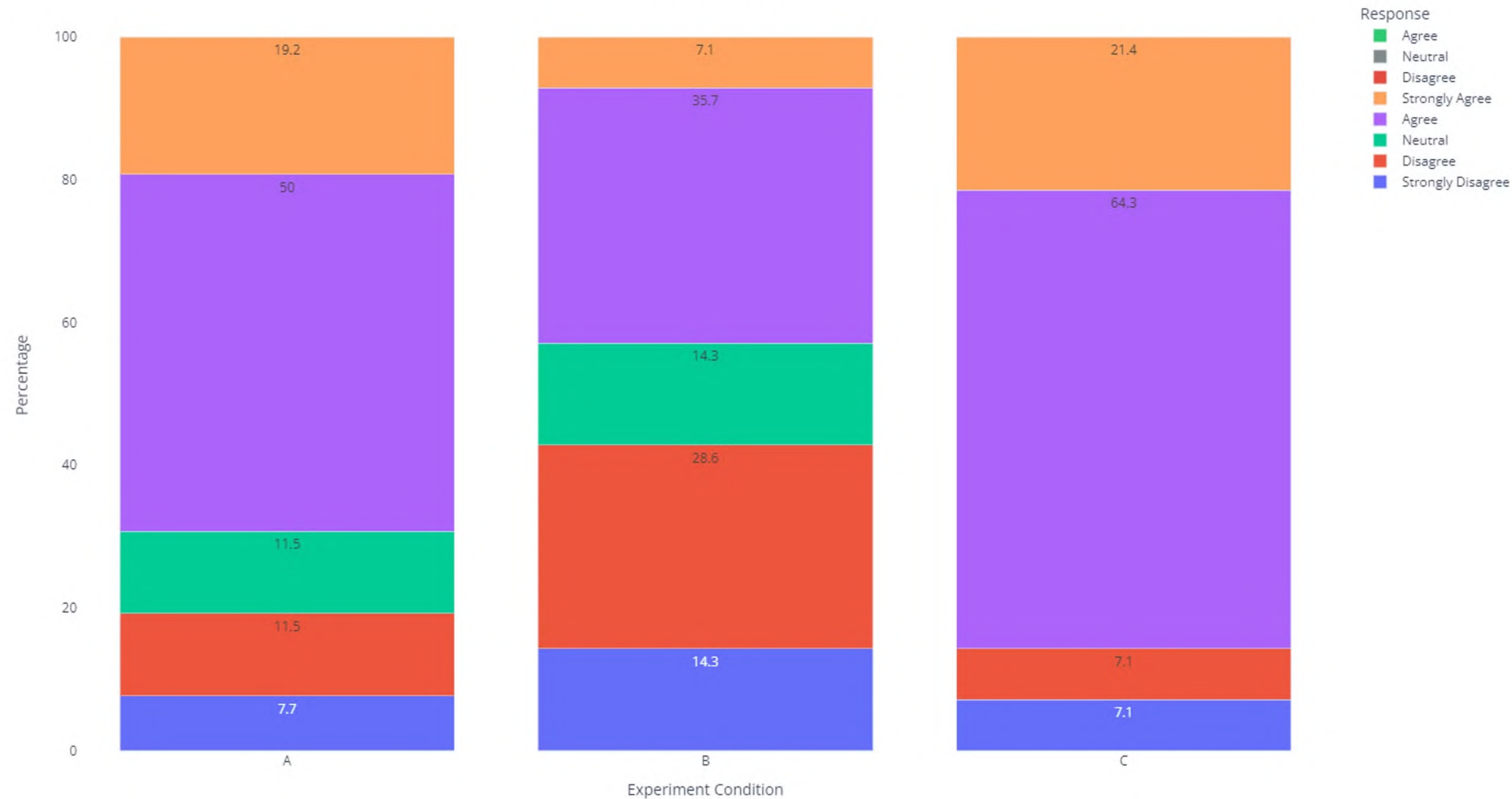
	mean	median	std	min	max	count
condition						
Functional UI	1.115385	1.2	0.582884	0.0	2.0	26
No UI	-0.085714	-0.4	1.036902	-1.4	1.8	14
Placebo UI	0.728571	1.2	0.994159	-1.4	2.0	14

Satisfying Metrics:

	mean	median	std	min	max	count
condition						
Functional UI	1.192308	1.25	0.704655	-0.50	2.0	26
No UI	0.035714	-0.50	1.208373	-1.75	2.0	14
Placebo UI	0.910714	1.25	0.978653	-0.75	2.0	14

Trust in Autonomous Vehicle Safety and Effectiveness

Stacked Bar Chart



Functional UI (A)

19.2% Strongly Agree

50% Agree

No UI (B)

7.1% Strongly Agree

35.7% Agree

Placebo UI (C)

21.4% Strongly Agree

64.3% Agree

6. CONCLUSION

*Option to Customize in Automated Vehicles provide a better **UX overall**, Usefulness, and Satisfaction however the differences are **not highly** significant. **Trust** in the proposed system was higher in percentages with Placebo UI condition.*

Personality traits Openness and Neuroticism play opposite roles in personalizing automated vehicle behavior.

01

DISCOVER

UNDERSTANDING THE LITERATURE

02

DEFINE

CONCEPTUALIZING & SETTING THE CONTEXT

03

DEVELOP

DEVELOPING THE SIMULATION - STUDY DESIGN

04

ANALYZE

DELIVERING FINAL PRODUCT & VALIDATING, REFINING SOLUTION

05

TRAIN MODEL

TRAINING A MACHINE LEARNING MODEL BASED ON COLLECTED DATA

LIMITATION

- **Motion Platform**
- **Sample Size**
- **Simulation Environment**

OUTLOOK

- **Second Round Experiment**
- **Improved UI**
- **User Centered Design**

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